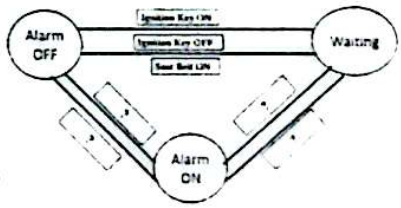
International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE Semester Final Examination, Spring-2025
Course Code: CSE-3523 **Course Title**. Microprocessors, Microcontrollers and Embedded Systems
Time: 2 hours 30 minutes **Total Marks: 50**

	Group-A	M	C
1a	Differentiate between maskable and non-maskable interrupts in 8086 with real-life examples.	3	2
b	Explain how a fixed interrupt vector can be generated using a DIP switch and 74ALS244 buffer during INTR in 8086. Include a neat and labeled circuit diagram.	4	2
c	Describe the sequence of operations performed by 8086 when a divide-by-zero operation occurs. Include the register changes and use of the stack during Type 0 interrupt.	3	2
	OR		
1. a	Suppose an external device generates an interrupt with vector number 60H. Describe how the CPU handles this interrupt after receiving the request. Including the steps involved in locating and executing the corresponding ISR.	5	2
b	Discuss the significance of the lower 3 interrupt types (Type 0–2) in 8086. Mention how each is triggered and handled.	5	2
2. Read the passage and answer the questions 2(a–c): Design a smart traffic signal system for a city that processes sensor inputs, prioritizes emergency vehicles, and reduces congestion using an embedded unit with multitasking, real-time interrupts, and optimized processing.			
a	Identify and justify 3 key processor features for the system. How does dual-core architecture improve multitasking?	3	2
b	Based on this scenario, suggest which type of processor (e.g., embedded, dual-core, arithmetic, bit-slice) would best suit the system. Explain your choice with architectural and application justification.	4	2
c	Explain how the use of cache memory and pipelining can enhance the performance of the traffic control system.	3	2

	Group-B		
3 a	What are the differences between isolated I/O and memory-mapped I/O? Discuss their advantages and disadvantages with relevant instruction examples.	3	3
b	Describe the internal operation of a basic input interface using a tri-state buffer. How does the SEL (Select) signal help in transferring data from input switches to the CPU? Illustrate with a labeled diagram.	3	3
c	What is handshaking in microprocessor I/O systems? Explain Mode 1 input handshaking of 82C55 with STB, IBF, INTR, and RD signals, including a timing diagram.	4	3
4a	Differentiate microprocessor and microcontroller by architecture, functionality, and application, with one example each.	3	3
b	8051 microcontroller comes in a 40-pin DIP package. Draw and explain the pin diagram of the 8051, identifying the purpose of the following pins: 1.EA (External Access) 2. PSEN (Program Store Enable) 3.ALE (Address Latch Enable) 4.RST (Reset) 5. Port 0 and Port 3 special functions.	4	3
c	Write an 8051 assembly language program to multiply two numbers stored at memory locations 40H and 41H. Store the result in registers A and B.	3	3
	OR		
4a	Write a program to add the values stored at memory locations 40H and 41H, and store the result at 50H and 51H.	4	3
b	Explain how a DC motor can be controlled using pin P1.3 and P1.4 of the 8051 microcontroller. Why is L293D necessary, and how do direction control signals work? Include circuit and code	6	3
5. a	Look at the figure of FSM(Seat Belt Warning System) at right and answer questions (a-c): Identify all states and events in the FSM. Explain each state in the context of a car's seat belt warning system, complete missing transitions, and explain the FSM.	3	3
b	Describe two conditions triggering transition from the 'Waiting' state and their effects. Provide another FSM example, e.g., a traffic light controller.	2	3
c	Identify three limitations of traditional embedded design and their impact on performance. Explain co-design, its benefits, and give an RTOS-based example.	5	3



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Department of Computer Science & Engineering

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B. Sc. in CSE Semester Final Examination, Autumn-2024

Course Code: CSE-3523 Course Title: Microprocessors, Microcontrollers and Embedded Systems

Total marks: *50* Time: *2* hours *30* minutes

Answer all the questions. Figures in the right hand margin indicate full marks. M:Marks, C:CLO, D: Domain

		M	C	D
	Group-A			
1	What is an interrupt? Describe the types of interrupts with examples.	3	1	U
2	Suppose you have four input devices: Disk, Printer, Reader, and Keyboard. These devices generate interrupts and are arranged in ascending order of priority based on their names. How does the Microprocessor 8086 handle these interrupts in a daisy-chain process? Explain the handling mechanism with a proper diagram.	5	1	An
3	What is the Interrupt Vector Table in the 8086 microprocessor? If the table occupies 1 KB of memory, calculate the total number of interrupt vectors it can hold. Explain the structure and purpose of each interrupt vector.	2	3	An
	OR,			
	Differentiate between hardware interrupts and software interrupts. Provide short notes on at least two software interrupts.			
4	What are Real Mode Interrupts and Protected Mode Interrupts? Compare both modes and explain their differences with examples.	4	2	U
	OR,			
	What is multitasking, and how is it implemented in processors?			
5	Draw and explain a simple method for generating the interrupt vector type number FFH in response to the INTR signal in the 8086 microprocessor. Provide a clear and labeled diagram.	2	1	U
6	How can the 8259A Programmable Interrupt Controller (PIC) be expanded to handle up to 64 interrupt requests without additional hardware? Explain the role of the master 8259A and the eight slave 8259A controllers in this configuration. Use a clear and labeled diagram to support your explanation.	4	3	An
	OR,			
	Explain the working method of a priority-based daisy chain mechanism.			
	Group-B			
7a)	What are the fundamental differences between isolated I/O and memory-mapped I/O methods, and how do these two approaches impact the interaction between a microprocessor and external devices in a computer system?	3	3	An
7b	Define handshaking. Explain the handshaking process with a proper example.	3	3	An
	OR,			
	What is Mode 1 operation of the 82C55?			
8	Describe the basic input and output operations within a microprocessor system. Provide a diagram to explain the data flow between the microprocessor and I/O devices.	4	1	U
	OR,			
	How does a 3-to-8 line decoder function in terms of input-output mapping, and what are its practical applications in digital circuits? Analyze its working mechanism			
9a)	What are the key differences between a microprocessor and a microcontroller in terms of architecture, functionality, and application? Discuss their distinct roles in embedded systems, including the components integrated within each, such as memory, I/O ports, and processing units, and explain how these differences affect their use in various applications like computing systems versus embedded devices.	4	2	U
9b	Describe the architecture of the 8051 microcontroller in detail, focusing on the key components such as the ALU, registers, control unit, timers, interrupt handling, and the special function registers. Explain the memory organization, including RAM, ROM, and external memory interfacing, and the role of the 8051's I/O ports.	3	3	An
	OR,			
	Analyze the hardware and flowchart design of a burglar alarm system. How does the integration of sensors, control logic, and output devices contribute to its functionality? Discuss the sequential flow of operations as depicted in the flowchart.			
10	How can multiplication and division operations be performed in the 8051 microcontroller? Explain the process.	3	2	An
11a)	Draw and explain the Embedded Systems Model. Label all key components, including input variables, output variables, user interface, embedded computer (software and hardware), and links to other systems.	2	1	U
11b)	Draw and explain the Hardware and Software Traditional Design Flow Diagram and the Co-Design Flow Diagram. Compare the two approaches, highlighting their differences. Discuss why the Co-Design approach is considered better than the Traditional Design Flow.	3	2	U
12)	How do embedded systems demonstrate behavior through finite state machine (FSM) models? Explain with a scenario-based example, such as a traffic light control system.	5	3	An

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International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc in CSE Final Examination, Spring-2024
Course Code: CSE-3523
Course Title: Microprocessors, Microcontrollers & Embedded Systems
Time: 2 hours 30 minutes **Total Marks:** 50

Answer all Five of the following questions. The figures in the right-hand margin indicate full marks.

Group - A

		M	C	P	D
1.a)	Describe how you would configure 74ALS244 to apply the interrupt vector type number B7H to the data bus in response to an interrupt request (INTR). Include the necessary connections and describe it.	4	3	2	A
b)	What are the practical advantages and challenges of expanding the number of interrupt request lines from one to seven in a Microcontrollers system? How does this impact the overall system's interrupt handling efficiency?	4	3	2	A
c)	Describe the role of the Non-Maskable Interrupt (NMI) in a microprocessor system. Why is it critical for handling urgent tasks?	2	3	2	U

OR

1. Read the passage and answer questions 1(a-b):
Suppose you have 4 input devices named Disk, Printer, Reader, and Keyboard. All devices are sending interrupts to the Microprocessor 8086.
N.B: Device names are listed in descending order of priority.

a)	What is an interrupt? Describe the types of interrupts.	3	3	2	A
b)	How does Microprocessor 8086 handle the above situation in the daisy chain process? Explain the handling process with the proper figure.	7	3	2	A
2.a)	Explain how the 8259A PIC can be expanded to handle up to 64 interrupt requests without additional hardware. What is the role of the master 8259A and the eight slave 8259A controllers in this configuration?	4	3	2	A
b)	Describe the role and operation of an interrupt vector table in a Microcontrollers system. How does it manage multiple interrupts efficiently?	4	3	2	U
c)	Explain the role of cache memory in microprocessors. How do L1, L2, and L3 cache levels differ in terms of performance and usage?	2	1	1	R

Group - B

3 a)	Write the differences between IO-mapped I/O and Memory-mapped I/O.	3	3	2	A

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b)	Draw the block diagram of 8255A.	3	3	2	
c)	How can you perform Multiplication and Division operations in Microcontroller 8051? Explain the process.	4	1	1	R

OR

3 a)	What role does a Programmable Logic Device (PLD) play in decoding I/O addresses in embedded systems? How does incorporating eight additional address lines (A15–A8) influence the address decoding process, and how does the PLD produce address strobes for I/O ports?	4	3	2	A
b)	Describe the basic input and output operations within a microprocessor system. Provide a diagram to explain the data movement between the microprocessor and I/O devices.	4	3	2	N
c)	Describe the basic Direct Memory Access (DMA) cycle in microprocessor systems. How does DMA facilitate data transfer between memory and peripheral devices without involving the CPU? Explain the key steps involved in a typical DMA transfer cycle and discuss the advantages of using DMA in data-intensive applications.	2	1	1	R
4.a)	Examine the pinout diagram of the Intel 8051 microcontroller. Describe the roles of Port 0, Port 1, Port 2, and Port 3, and explain how each port enables the microcontroller to interface with external memory, input/output devices, and other peripherals.	4	2	1	A
b)	Explain how zones 1, 2, 3, and 4 are activated in an 8051 microcontroller program that displays different zone numbers on a seven-segment display. Include the logic used to determine the active zone and how this information is output to the display.	4	2	1	N
c)	Explain the key components typically integrated into a System on Chip (SoC). How do these components work together to provide a complete solution for embedded systems, and what challenges do designers face when developing SoC architectures?	2	2	1	R
5 a)	How does the handshaking or polling method synchronize the communication between a microprocessor and slower I/O devices, such as a parallel printer? Provide an example and explain the mechanisms used to slow down the microprocessor to match the speed of the printer.	3	3	2	A
b)	Develop a finite state machine(FSM) for an elevator system that operates in a multi-story building. The FSM should manage the elevator's movements, door operations, and respond to user inputs such as floor selection and door open/close commands. Explain the states and transitions involved, including how the system prioritizes requests from different floors.	4	2	1	R
c)	What is hardware-software codesign, and how does it enhance the development of embedded systems? Discuss the key benefits and challenges associated with integrating hardware and software design processes in the development lifecycle.	3	2	1	U

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Department of Computer Science and Engineering
B. Sc. in CSE
Final Exam, Autumn 2023
Course Code: CSE 3525 Course Title: Microprocessors, Microcontrollers & Embedded Systems Full Marks: 50
Time: 2 hours 30 minutes

[Answer all the questions, Figures in the right hand margin indicate full marks]

		Group A			
1.	a)	What is an interrupt? Describe the types of interrupts with examples.	CO3	Un	5
1.	b)	Suppose you have four input devices: Disk, Printer, Reader, and Keyboard. All these devices are sending interrupts to the Microprocessor 8086. Note that the device names are listed in descending order of priority.. How does Microprocessor 8086 handle the above situation in the daisy chain process? Explain the handling process with the proper figure.	CO3	An	5
		OR			
1.	b)	What role does the 8259A Programmable Interrupt Controller (PIC) play in managing interrupt priorities and vectoring in microprocessor systems, and how does the expand-ability of the 8259A to accept up to 64 interrupt requests impact the scalability and performance of these systems?	CO3	An	5
2.	a)	What is the significance of using interrupts in the context of keyboard input, and how does the generation of a single pulse by the keyboard encode of the system?Explain with timeline diagram	CO1	Un	5
2.	b)	How does the NMI function in terms of its triggering conditions and its role in detecting critical events such as power failures and parity errors in microprocessor-based systems, and why is it essential for maintaining system reliability?	CO1	Un	5
		OR			
2.	b)	What is bit slicing in microprocessor design, and how does this approach enable the construction of specialized, high-performance processors for specific applications or functions by combining multiple bit-slice units?	CO1	Un	5
		Group B			
3.	a)	What are the fundamental differences between isolated I/O and memory-mapped I/O methods, and how do these two approaches affect the interaction between a microprocessor and external devices in a computer system?.	CO3	Un	5
3.	b)	How can you perform Multiplication and Division operations in Microcontroller 8051? Explain the process.	CO3	Un	5
		OR			
3.	b)	What are the primary components involved in basic input and output operations within a microprocessor system, and how are these operations defined in terms of IN and OUT, referring to data movement between the microprocessor and I/O devices?Explain with proper diagram.	CO3	Un	5
4.	a)	Recently, Amit and Arafat learned about the 8051 microcontroller. However, they are aware that there is a pin that separates address and data lines and another pin that disables internal ROM, but they are having trouble recognizing them. Help them by describing these pins.	CO2	Ap	5
4.	b)	What are the key functions of the main pins in the Intel 8051 microcontroller, as depicted in the pinout diagram, and how do these pins contribute to the microcontroller's operation and its ability to interface with external devices?	CO2	Ap	5
5.	a)	What is a System-on-a-Chip (SoC)? Describe each of its components with a diagram.	CO3	An	5
5.	b)	What is the significance of the PLD (Programmable Logic Device) in the context of decoding I/O addresses in embedded systems? How does the addition of eight address lines (A15–A8) affect the decoding process, and how does the PLD generate address strobes for I/O ports?	CO3	Un	5
		OR			
5.	b)	Provide a brief explanation of modeling embedded system behavior, illustrated with an example of a finite state machine model.	CO3	Un	5

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International Islamic University Chittagong

Department of Computer Science & Engineering (CSE)
Final Examination, Spring-2023
Course Code: CSE 3523
Course Title: Microprocessors, Microcontrollers and Embedded Systems
Total Marks: 50 **Time: 2 Hour 30 minutes**
[Answer three of the following questions. Figures in the right-hand margin indicate full marks.]

GROUP-A

1.(a)	Explain with figure how interrupts inputs are expanded with using 74ALS244 three state buffer. **Or,** Explain how microprocessor can program with 8259A and also explain function of the pin.	4	CLO2
1.(b)	Explain interrupt vector and its table with figure. What is interrupt service procedure? **Or,** Draw the circuit diagram for generating interrupts vector type number 1Ah in response to INTA.	4	CLO3
1.(c)	Show a timeline that indicates interrupt usage in a typical system.	2	CLO3
2.(a)	Why and how daisy chain interrupt is better than using 74ALS244 explain with proper diagram.	4	CLO3
2.(b)	Define the following terms: a) Bit slicing b) Multitasking c)Threads d)Core 2 Processors	4	CLO1
2.(c)	Write short note on INT and INTA. **Or,** Describe the function of Slave program/enable buffer pin.	2	CLO1

GROUP-B

3.(a)	Write the differences between IO-mapped I/O and Memory-mapped I/O	3	CLO1
3.(b)	Drawtheblockdiagramof8255A **Or,** Draw the block diagram of the Intel 80286 Microprocessor	3	CLO1
3.(c)	How can you perform Multiplication and Division operations in Microcontroller8051? Explain the process. **Or,** Differentiate between microprocessor and microcontroller with proper example.	4	CLO2
4	Recently, Amit and Arafat have learned about Microcontroller 8051. However, they know that there is a pin that separates address and data lines and another pin that discards internal ROM but the y is not able to recognize them. Help them by answering the following questions 4(a-b).		
4(a)	What are those two pins that are mentioned in the above passage? Briefly explain them.	7	CLO2
4(b)	Give them a visual representation of Microcontroller 8051 by drawing a block diagram.	3	CLO2
5(a)	What are the characteristics of an embedded system? What do you mean by hardware and software co-design?	4	CLO1
5(b)	Briefly explain the modeling of embedded system behavior with a finite state machine model.	3	CLO1
5(c)	What is a system on the chip? Explain with a diagram. **Or** Describe the components of Embedded system	3	CLO1

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International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE, Semester Final Exam, Spring 2022
Course Code: CSE 3523 **Course Title:** Microprocessors, Microcontrollers & Embedded Systems
Total Marks: 50 **Time:** 2.5 Hours
[Answer *all the five* questions. Figures in the right-hand margin indicate full marks.]

Course Outcomes and Bloom's Levels are mentioned in additional Columns

		CLO	Mark
	PART A		
1(a)	In the early years of computing, the processor has to waste a number of clock cycles just for checking the signal in the system, by this processor will become busy unnecessarily. If any signal came for the process, the processor will take some time to process the signal due to the checking which is known as the polling method. So system performance also will be degraded and the response time of the system will also decrease. Suggest a better process to handle the signal. Explain a general process of it.	3	5
(b)	Can your suggested process in 1(a) be handled in parallel? Explain it.	3	5
	OR		
	How interrupt vector type number 80H is applied to the data bus in response to an INTR using three state buffer.		
2.(a)	Write short note on: i. L1 and L2 Cache, ii.BOUND	1	4
(b)	Draw the block diagram of 8255A.	1	4
	OR		
	Draw and describe the pin out diagram of 8259A programmable interrupt controller.		
(c)	What are the difference between hardware interrupt and software interrupt.	2	2
	PART B		
3.(a)	Suppose you are building an hardware project named digital alarm clock. Will you use Microprocessor or Microcontroller? Explain the reasons behind it using diagrams.	4	3
(b)	Explain the basic DMA operation with timing diagram	4	4
(c)	Write the differences between IO-mapped I/O and Memory-mapped I/O.	4	3
4.(a)	Recently, Amit and Arafat learned about Microcontroller 8051. However, they are unsure of how to access RAM or ROM due to the fact that the device contains both internal and external RAM and ROM. Help them by answering the following questions – i. To interface a 16-bit Memory what port(s) they can use? Describe the port(s). ii. How can they access external RAM with 8 and 16 bits of addressing?	2	6
	OR		
	Design a burglar alarm system using 8051 microcontrollers. When a burglar will come to your house then alarm sounds are generated. Draw the circuit diagram of burglar alarm system where port 1 is used as input and port 3 is used as output port. Also draw the flowchart when zone 2 is activated.		
(b)	Describe the architecture of 8051 micro-controller with figure.	1	4
5.(a)	Explain why hardware-software co-design is better than traditional design. Describe it with proper diagram.	1	4
	OR		
	How can you optimize an embedded system?		
(b)	Draw a general block diagram for an embedded system.	1	3
(c)	Explain interrupt latency and how can we decrease it	1	3

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